

Lab 5: Bipolar Junction Transistors

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This lab investigates the behavior of bipolar junction transistors. A bipolar junction transistor (BJT) is a semiconductor primarily used for amplifying signals where a small base current controls a collector current to amplify signal strength. They can also be used to act as a switch with the same process. They consist of three terminals and are formed from two p-n junctions allowing for the creation of either an NPN or PNP structure. The 2N3904, is the specific BJT used in this lab, is an NPN device. The emitter is heavily doped with n-type material, the base is slightly doped with p-type material, and the collector is moderately doped with n-type material. The device functions by controlling the carriers by using a small input at the base terminal. In an NPN transistor, like in this lab, A forward-biased base-emitter junction allows electrons to move from the emitter to the base. A reverse-biased base-emitter junction instead takes those electrons and moves the majority of them into the collector. This is what creates a current gain, taking a small base current producing an increased collector current.

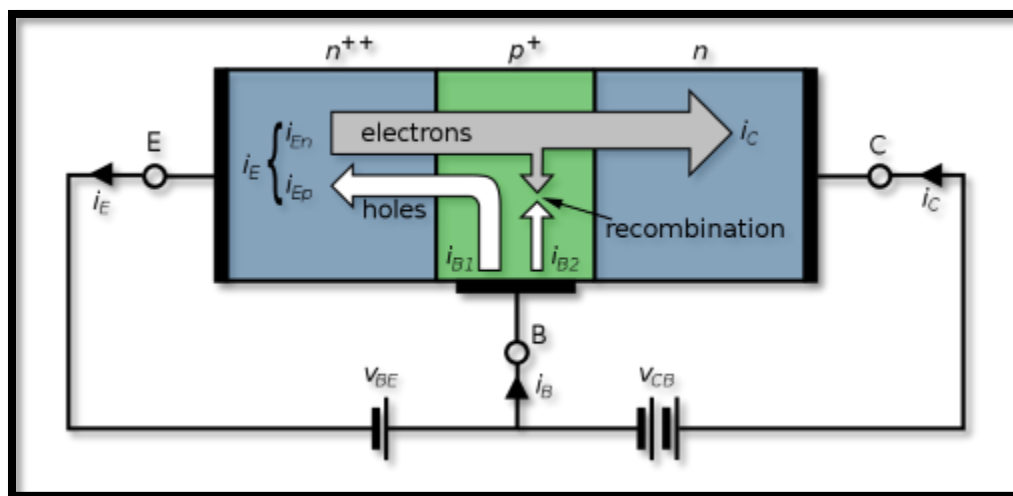


Figure 1: Structure/Function of a BJT

Fundamentals:

As seen in the image of a BJT's structure above, a standard silicon NPN BJT is made of a p-type region, the base, in the middle between two n-type regions, the emitter and collector. The emitter is the heaviest doped of the regions injected with a large number of carriers. The base is thinner and less doped allowing the carriers to move across. The collector is doped and usually larger so it can collect the injected carriers. For a PNP BJT, the polarities are reversed. There is a p-type emitter, n-type based, and a p-type collector. The overall structure and process of operation are the same for both, but the carrier type and current directions flip depending on the device.

There are four different primary modes of operation for a BJT depending on how it is biased. The first is cutoff and is when both the base-emitter and base-collector junctions are reverse biased. There is no major carrier injection meaning that I_B , I_C , and I_E , are zero. The BJT behaves like a short/open circuit. The next is forward active and is when the base-emitter is forward biased while the base-collector junction is reverse biased. Electrons are sent into the emitter and cross the base where they are funneled into the collector. This creates the collector current, $I_C = \beta I_B$, where amplification usually happens. Saturation is the third primary mode of operation. It occurs when the base-emitter and base-collector junctions are forward biased. This means that the collector is not able to collect carriers as well as before. So I_C is limited and the BJT behaves like a closed switch with a small saturation voltage over the collector-emitter junction. The last primary mode of operation is known as reverse active. It occurs when the base-emitter junction is reverse biased while the base-collector junction is forward biased. The transistor should still operate, but with far worse performance since the collector is now acting as the emitter and cannot inject carriers as efficiently.

The two significant applications for a BJT are to act as a gate and as an amplifier. When acting as a gate the BJT switches between cutoff and saturation mode. Without a base current, the base-emitter junction will stay in reverse bias and the device will keep blocking current. When enough of a base current is applied to the base-emitter junction, it will be forward biased and enter saturation mode and allow current to flow. When acting as an amplifier the BJT must be in forward active mode (base-emitter forward biased and base-collector reverse biased.). It will produce a collector current proportional to the base current ($I_C = \beta I_B$). Even a small change in the base current will generate a greater collector current. The collector converting the current change into a voltage change is what results in a massive, amplified voltage output.

Experiment 1: Demonstrate the Regimes of Operation of a BJT

The goal of experiment 1 is to measure the different operating modes of a BJT. A 2N3904 NPN transistor is used. Values are found for V_B , V_C , and V_E . Three currents I_B , I_C , and I_E , are recorded. A Keysight U2722A SMU and its AMM software is utilized to record measurements. The setup is simple. The transistor is connected to a breadboard and wired to the SMU. Values are set in the software corresponding to each operating mode the transistor can be in. The current are then measured and displayed in the table below:

Mode/Pin	Saturation	Reverse Active	Cut Off	Forward Active
Base	0.7 V and 0.04 mA	1 V and 5.05 mA	1 V and 0 mA	0.7 V and 0.5 mA
Collector	0.5 V and 8.6 mA	0.7 V and 0.92 mA	0.7 V and 0 mA	1 V and 10 mA
Emitter	0V and 9mA	0 V and 5.97 mA	0 V and 0 mA	0 V and 10.5 mA

The value for the amplification factor, β , is calculated for the forward active region.

$$I_B = 0.5 \text{ mA}, I_C = 10 \text{ mA}$$

$$\beta = \frac{I_C}{I_B} = \frac{10 \text{ mA}}{0.5 \text{ mA}} = 20$$

The datasheet expresses the value with a range of 40-300. Our calculated value is significantly lower than even the minimum given by the datasheet. I suspect that during our recorded measurements, the voltages should have been set higher to ensure the transistor entered a strong forward active mode. Although lower than the datasheet, the transistor still behaved as suspected. With higher voltages ensuring forward active mode activation the gain factor should be higher.

Concept Question(s):

According to the recorded data, each region is relatively ideal except for the forward active. The saturation is acceptable but V_C should have been lower, the reverse active is the most ideal, the cutoff mode works as intended but V_B should have been closer to the threshold of 0.6-0.7V, and the forward active is the least ideal as the gain factor is too low. Although each region behaves as expected, with better chosen measurement values the recorded behavior of each mode can be more defined.

Experiment 2: Demonstrating the Forward Active Mode of a BJT

The goal of experiment 2 is to understand the forward active mode for an NPN BJT. The SMU is used again to power the 2N3904 in the common emitter configuration. Voltages are applied to V_C and V_E and I_B . A sweep is run to generate a series of curves that is typical for this device in

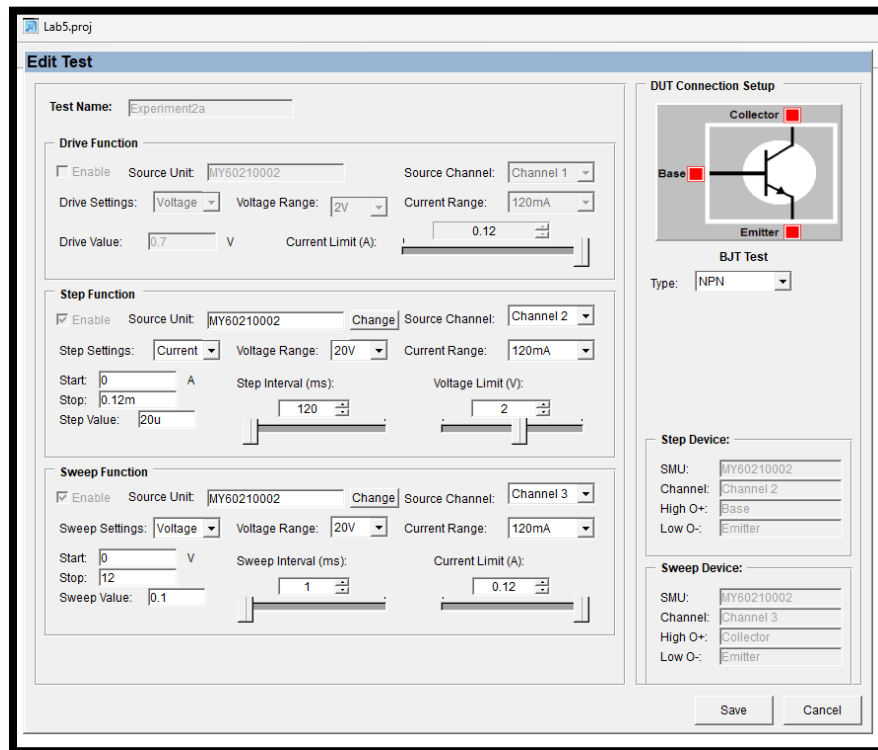


Figure 2: Experiment Setup Configuration

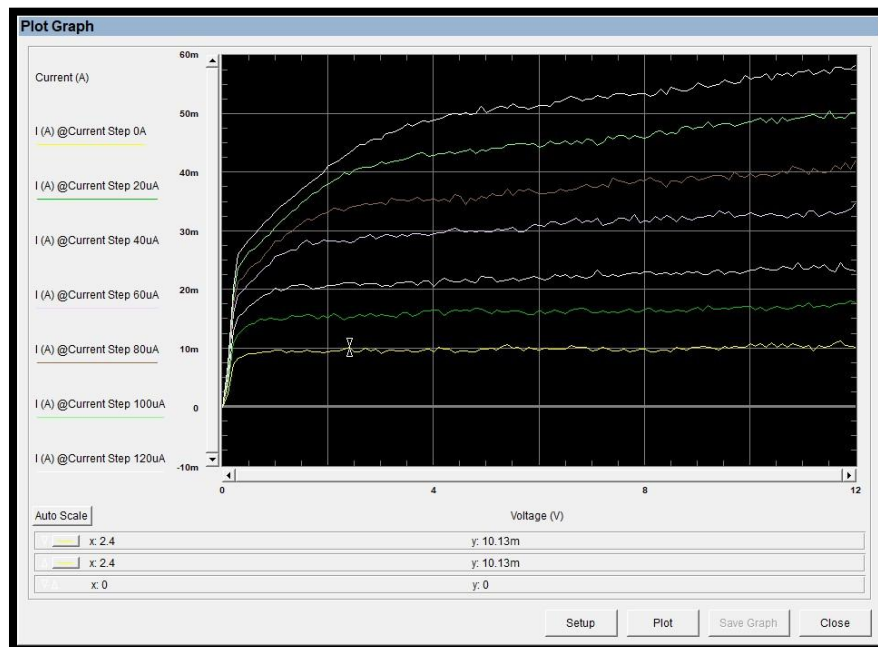


Figure 3: Generated Series of Curves in Forward Active

For $I_B = 100 \mu\text{A}$, a plot of the gain coefficient, β , is created as a function of V_{CE} .

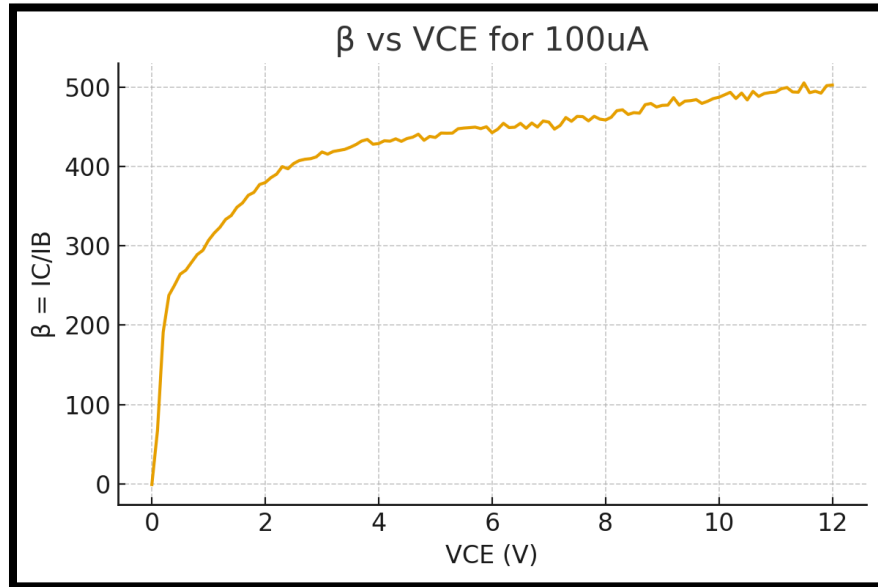


Figure 4: Plot of β as function of V_{CE}

Concept Question(s):

The curves are flat in the active region and lower at a low V_{CE} which matches the expected behavior. The noise seen is most likely from the SMU itself and breadboard wiring. If the curves were to be measured over a longer period of time it would not be surprising to see each curve start to droop due to an increase of device temperature. The measured β values of ~300-500 match well with the datasheet. They well exceed the minimum but near the end remain on the slightly higher side. I believe this small discrepancy can be corrected using better measurement scaling or adjusting our current step in our experiment's setup.

Experiment 3: Switching a Device On and Off with a BJT

The goal of experiment 3 is to make the NPN BJT behave as a switching circuit by turning on and off an LED. The setup involved recreating a circuit for an LED's current to be measured. An oscilloscope is used to capture the voltage across the LED and resistor. These values are used to calculate the current running through the LED in the on and off conditions of the BJT as well as power consumption in the control line.

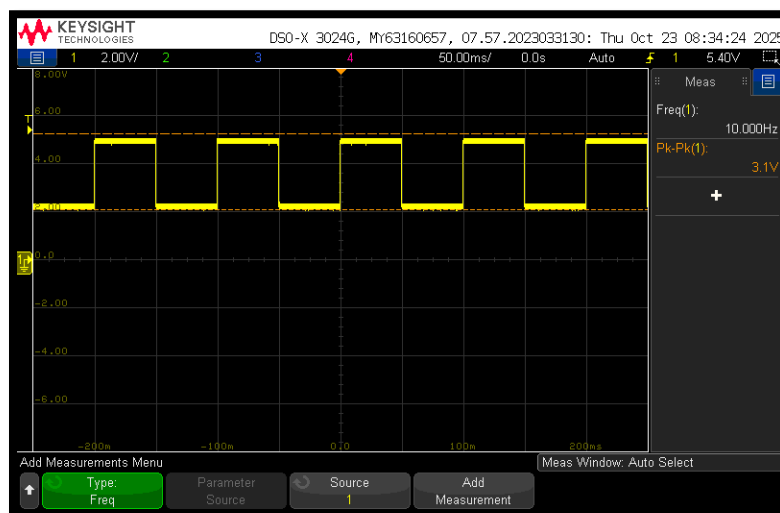


Figure 5: Voltage across LED

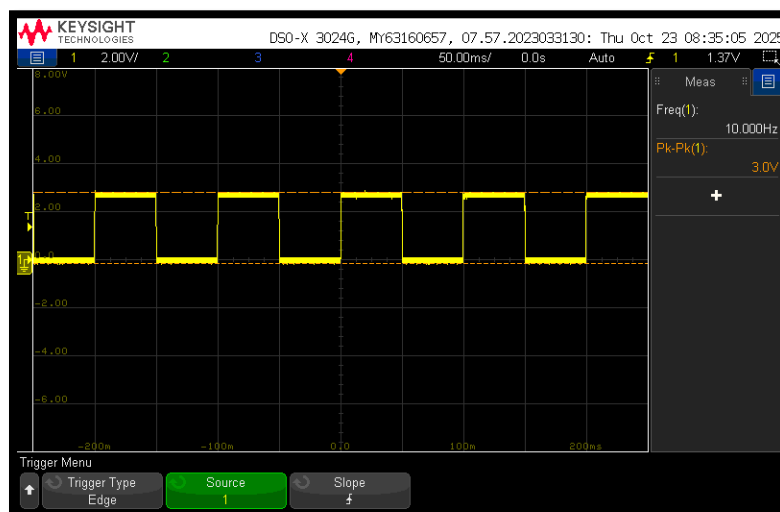


Figure 6: Voltage across resistor

The following measurements and calculations are recorded:

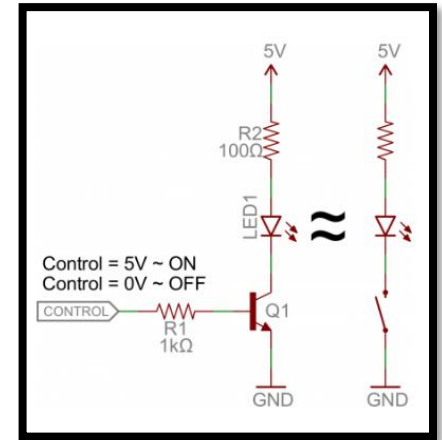
$$V_{LED} = 3.1 \text{ V}$$

$$V_R = 3.0 \text{ V}$$

$$V_{CE (saturation)} = 5 \text{ V} - 3.1 \text{ V} - 3.0 \text{ V} \approx 0 \text{ V}$$

$$I_{LED \text{ ON}} = \frac{V_R}{R_2} = \frac{3.0 \text{ V}}{100 \Omega} \approx 30 \text{ mA}$$

$$I_{LED \text{ OFF}} = 0 \text{ mA}$$



Verifying Kirchoff's Law:

Figure 7: Recreated LED Circuit

$$I_B = \frac{5 \text{ V} - 0.7 \text{ V}}{1 \text{ k}\Omega} \approx 4.3 \text{ mA}$$

$$I_C = I_{LED \text{ ON}} = 30 \text{ mA}$$

KCL at transistor:

$$I_E = I_C + I_B = 30 \text{ mA} + 4.3 \text{ mA} = 34.3 \text{ mA}$$

KCL holds within reason of the scope readings. The forced saturation is:

$$\frac{I_C}{I_B} = \frac{30 \text{ mA}}{4.3 \text{ mA}} \approx 7$$

This is low as expected to guarantee an enabled saturation mode.

The power consumption in the control line:

$$P_{Control} = V_{in} I_B = 5 \text{ V} \times 4.3 \text{ mA} \approx 21.5 \text{ mW}$$

The input waveform being used is a 10 Hz, 1,0 V_{pk-pk} square waveform with a 500-mV offset. This aligns with the observed behavior on the scope. The LED voltages steps between 0 V to 3.1 V perfectly in sync as well as over the 100 Ω resistor.

Concept Question(s):

Utilizing a BJT in this way lets a very small base current control a much larger LED current. This gives current amplification where a few source mA can manipulate tens of mA through the LED without stress. It allows for clean operation behavior of saturation vs cutoff and can also allow for more features such as LED dimming without changing the control source. The best way to reduce power consumption is to reduce the LED current. This can be done by increasing the series resistor or changing the input waveform to a lower duty cycle. The main limit to this power consumption challenge is the minimum required LED brightness and characteristics of the transistor itself. If a certain brightness must be met, you can only decrease the power consumption so much. If the transistor's characteristics for its different operation modes such as its saturation requires a large enough minimum V_{CE} and V_{BE} then that will limit what can be changed as well.

Experiment 4: Amplifying a signal with a BJT

The goal of experiment 4 is to make the NPN BJT behave as a simple amplifying circuit by showing that it can run a small speaker. The setup involved recreating a circuit the speaker and capturing the voltage. An oscilloscope is used for this to determine the gain factor with respect to the input signal. The recreated circuit is below:

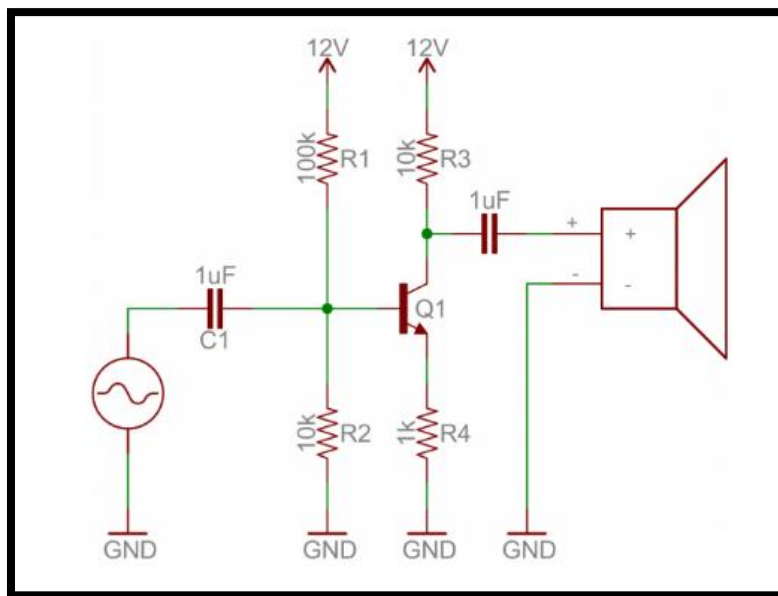


Figure 8: Recreated Speaker Circuit

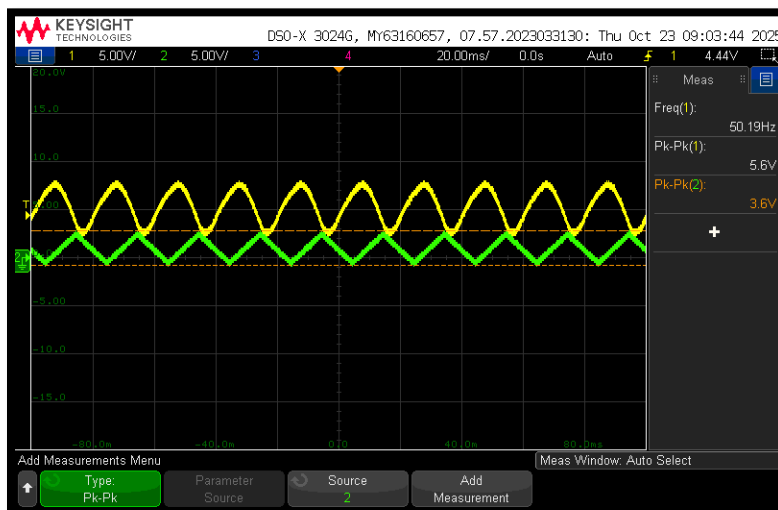


Figure 9: Ch 1: Output Waveform Ch 2: Input Waveform

The voltage gain factor is calculated:

$$A_v = \frac{V_{out, \text{Pk-Pk}}}{V_{in, \text{Pk-Pk}}} = \frac{5.6 \text{ V}}{3.6 \text{ V}} \approx 1.56$$

Concept Question(s):

The BJT is running in forward active mode. The base-emitter junction is forward biased by the DC voltage divider (R1 and R2) on its base and the collector-base junction is reverse biased by the collector resistor. The applied voltages are what set this. The base is fixed around 0.6 – 0.8 V above the emitter due to that DC bias voltage divider. The emitter resistor is what sets the current. The collector supply keeps the collector a few volts above the base ensuring reverse bias of the collector-base junction. This is what keeps the device operating in forward active mode allowing for amplification.

Simulation:

The simulation kept crashing when trying to change the values of the emitter, base, and collector regions. All I could get is the default simulation and its β vs I_C .

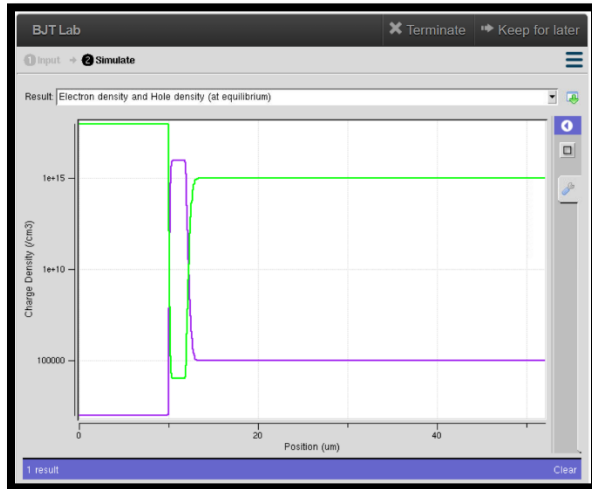


Figure 10: Default Simulation

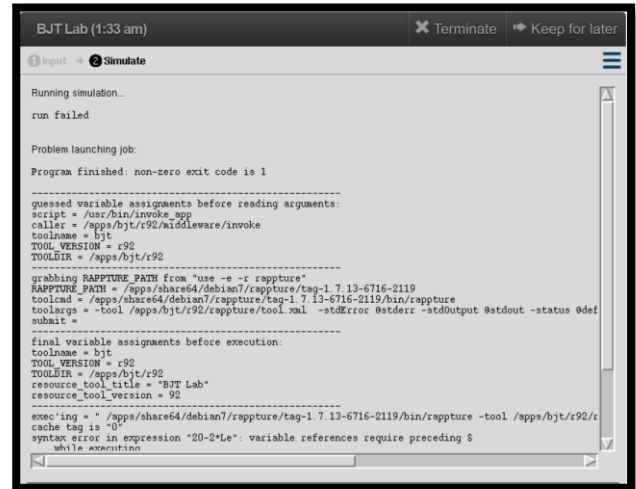


Figure 11: Adjusting Length Crash

What should be seen when changing the values of V_{BE} is a similar series of curves from experiment 2. A short base should make the curves sit higher (Bigger I_C for same V_{BE}) and flatter (smaller active region slope). Shrinking the base will cause fewer carries to move to the base increasing β . A higher gain should result in a higher I_C and flatter curves than the default parameters.

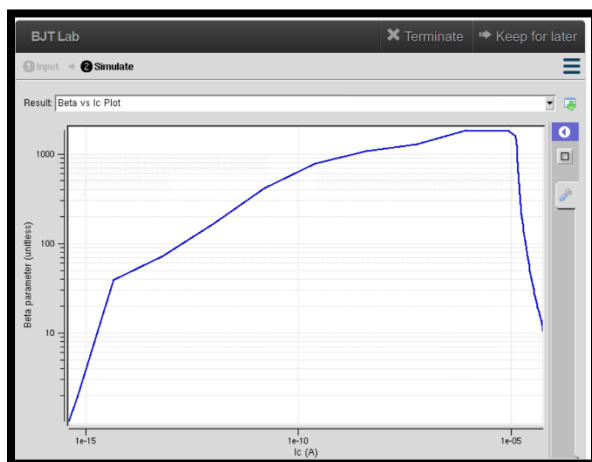


Figure 12: Default Simulation β vs I_C

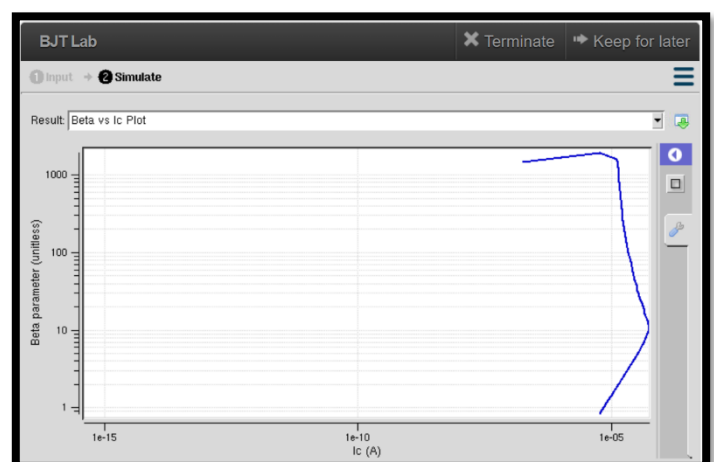


Figure 13: $V_{BE\text{ Start}} = 1V$, $V_{BE\text{ End}} = 3V$ β vs I_C

The default simulation shows a normal curve rising from small currents, peaking around a moderate I_C , and then steeply falling off for a higher I_C . By shrinking the base length, the curve should shift upward and the peak should move slightly further to the right near a higher I_C .

Conclusion:

This lab taught me how a BJT behaves and its extremely useful operation modes for its various applications. Over the four different experiments, the BJT's function was demonstrated as a controlled switch and small-signal amplifier. Experiment 1 served to investigate the four different operating modes of the device and the differences between them. Experiment 2 served to expand on the forward active mode showing on collector current and gain depend on V_{CE} . Experiment 3 served to illustrate how a BJT can be useful when acting as a switch using a small base current to control an LED's current. Experiment 4 served to investigate the small-signal voltage amplification the device is capable of utilizing different biasing effects to hold the transistor in a forward active mode. Together, each experiment ties into the previous to provide an explanation of the practicality and function of an NPN BJT. I learned how to interpret small data points and refer back to the device's datasheet to validate results.

GenAI Usage:

AI was used to help generate figure 4 from the recorded .csv data file.

Citations:

BJT Structure Image:

“Bipolar Junction Transistors: What Are They & What Do They Do.” *Electromechanical*

Technician Training – George Brown College News Blog, 31 Jan. 2022,

www.emcourse.com/news-blog/bipolar-junction-transistors-what-are-they-what-do-they-do